Transcriptomic Lineage Partitioning Resolves Venetoclax Resistance in Acute Myeloid Leukemia

[Author Name], MD, PhD[1,*]

[1]Department of Hematology and Oncology, [Institution]

*Correspondence: [email]

Key Points

- Lineage partitioning identifies monocytic vs progenitor AML states that predict Venetoclax response independent of mutations.
- The 9-gene Venetoclax Response Score stratifies trial efficacy and exposes targetable collateral salvage sensitivities.

Abstract

Background: Venetoclax plus hypomethylating agents has transformed acute myeloid leukemia (AML) therapy, but predictive biomarkers are lacking. We hypothesized that transcriptomic lineage states dictate BCL-2 dependency and target response independent of somatic mutations.

Methods: Molecular subtypes were defined via consensus clustering in the BeatAML cohort ($N = 450$) and validated across five cohorts ($N = 2{,}567$). Predictive value was assessed using R^2 variance improvement models controlling for 11 clinical driver mutations across 155 ex vivo drug screens, matched proteomics, and single-cell CITE-seq.

Results: We identified two stable states: Cluster 1 (monocytic-primed; 40.9%) and Cluster 2 (primitive/progenitor-like; 59.1%). These states were not prognostic after adjusting for mutations ($p = 0.59$) but were highly predictive of Venetoclax response ($p = 6.37 \times 10^{-19}$), providing a +115.7% relative R^2 improvement over mutational models and outperforming LSC17 stemness scores ($p = 0.008$). Relapsed blasts selected for monocytic-primed states under therapy, shifting apoptotic dependencies from BCL2 to MCL1. This resistance axis was validated in an independent cohort ($N = 367$) and matched proteomics, exposing targetable salvage vulnerabilities to MEK, HDAC, and mTOR inhibition.

Conclusions: Transcriptomic lineage states define a mutation-independent axis of cell maturity that dictates targeted therapy response, providing a precision roadmap to bypass resistance.

1 Introduction

Acute myeloid leukemia (AML) is a highly aggressive and biologically heterogeneous hematologic malignancy characterized by the clonal expansion of immature myeloid blasts in the bone marrow and peripheral blood [1]. For over four decades, standard induction chemotherapy consisting of cytarabine and an anthracycline ("7+3") remained the mainstay of treatment, yielding 5-year overall survival rates of less than 30% in older adults [2, 3]. The recent clinical development and FDA approval of Venetoclax—a potent, highly selective oral small-molecule inhibitor of the anti-apoptotic protein BCL-2—in combination with hypomethylating agents (HMAs, such as Azacitidine or Decitabine) or low-dose cytarabine (LDAC), has fundamentally transformed the therapeutic landscape for older or unfit AML patients [4, 5, 6, 7]. In pivotal Phase III clinical trials, Venetoclax + Azacitidine achieved complete remission (CR) or CR with incomplete hematologic recovery (CRi) rates of 66%, alongside a statistically significant overall survival benefit compared to Azacitidine monotherapy [7].

Despite these groundbreaking clinical advances, primary treatment failure and secondary relapse under Venetoclax-based combinations remain pervasive clinical challenges [8, 9]. Approximately 20–30% of patients exhibit primary refractory disease, while the vast majority of initial responders eventually experience lethal disease recurrence [4]. Current clinical risk stratification in AML relies heavily on cytogenetic abnormalities and recurrent somatic driver mutations, as codified by the European LeukemiaNet (ELN) recommendations [2, 3, 10]. Somatic alterations in genes such as *NPM1*, *FLT3-ITD*, *DNMT3A*, and *TP53* provide critical prognostic information regarding overall survival under traditional intensive chemotherapy [11]. However, their utility as predictive biomarkers for targeted Venetoclax response is remarkably limited, with only *NPM1* and *IDH1/2* mutations showing modest prospective association with improved response, while *TP53* mutations confer resistance [8, 9].

The biological basis for these differential clinical responses remains partially understood. Crucially, emerging functional genomic and single-cell studies suggest that Venetoclax resistance frequently manifests through non-mutational mechanisms—specifically through cell-state transitions, developmental lineage selection, and anti-apoptotic dependency switching—rather than the acquisition of novel driver mutations [12, 13, 14, 15]. Leukemic blasts spanning the spectrum of myeloid differentiation display distinct bioenergetic profiles and anti-apoptotic dependencies: primitive stem/progenitor-like blasts rely predominantly on BCL-2 and oxidative phosphorylation, whereas mature monocytic-differentiated blasts switch to MCL-1 and BCL-XL survival dependency [15, 16, 17, 18]. However, a fundamental knowledge gap remains: whether transcriptomic lineage states can be systematically quantified to predict drug sensitivity independent of somatic driver mutations across large multi-cohort populations, and whether this lineage axis exposes actionable collateral vulnerabilities in Venetoclax-resistant AML.

To resolve this challenge, it is critical to determine whether transcriptomic developmental cell maturity states act as independent predictive biomarkers for targeted therapy. In high-impact clinical settings such as the landmark VIALE-A trial, patient outcomes varied considerably across genetic subgroups. While patients with nucleophosmin 1 (*NPM1*) mutations achieved high response rates and durable remissions under Venetoclax combinations, those with monocytic leukemia (FAB subtype M5) or tumor protein p53 (*TP53*) mutations experienced early relapse and treatment resistance. Developing a quantitative, continuous transcriptomic classifier that directly models these developmental cell-state transitions could bridge this clinical gap, allowing clinicians to stratify patients prospectively and identify rational, combination-based salvage therapies before clinical relapse occurs.

In this study, we hypothesize that transcriptomic lineage partitioning defines a fundamental, mutation-independent dimension of AML biology that dictates BCL-2 target dependency and global pharmacological sensitivity. Utilizing multi-omic profiling, functional genomics, and single-cell sequencing across 3,029 AML patients from eight independent discovery and validation cohorts spanning three continents, we demonstrate that transcriptomic lineage states predict Venetoclax response with extraordinary accuracy beyond mutational models. Furthermore, we derive and validate the Venetoclax Response Score (VRS) as a continuous bedside biomarker, characterize single-cell resistance dynamics, and identify targetable salvage vulnerabilities in Venetoclax-resistant AML.

2 Methods

2.1 Patient Cohorts and Multi-Omics Data Processing

This study integrated transcriptomic, genomic, functional drug profiling, proteomic, and clinical outcome data from ¿3,500 adult and pediatric AML patients across 8 independent cohorts: (1) BeatAML discovery cohort ($N = 671$ overall transcriptomic set; $N = 520$ drug-screened set; $N = 320$ gold-standard multi-omics cohort with matching RNA-seq, targeted DNA sequencing, ex vivo drug response, and clinical annotation) [12]; (2) TCGA-LAML adult validation cohort ($N = 151$) [11]; (3) BeatAML 2.0 independent validation cohort ($N = 367$) [30, 13]; (4) CPTAC BeatAML mass-spectrometry quantitative proteomics cohort ($N = 327$); (5) van Galen single-cell RNA-seq bone marrow hierarchy ($N = 16$ patients, 38,412 cells) [14]; (6) Pei et al. CITE-seq single-cell ADT surface protein cohort ($N = 4,426$ malignant blasts at diagnosis and relapse) [15]; (7) Cancer Dependency Map (DepMap) CRISPR-Cas9 essentiality screen ($N = 24$ myeloid leukemia cell lines); and (8) European multicenter trial cohorts ($N = 626$ total; German AMLCG GPL96 $N = 163$, AMLCG GPL570 $N = 79$, German-Austrian AMLSG $N = 211$, and GSE106291 chemotherapy induction $N = 235$).

Raw sequencing reads from RNA-seq were aligned to the human genome reference (GRCh38) using the STAR alignment algorithm, and gene expression was quantified using the RSEM software package to obtain raw counts. For mass-spectrometry quantitative proteomics, primary bone marrow mononuclear cells were lysed, digested with trypsin, and labeled with tandem mass tags (TMT). Labeled peptides were separated using high-pH reverse-phase liquid chromatography and analyzed on an Orbitrap Fusion Lumos mass spectrometer. Raw mass spectra were searched against the UniProt database to identify and quantify protein abundances. Raw gene expression counts were normalized, transformed, and batch-corrected across centers using ComBat empirical Bayes batch correction [19], preserving the clinical risk and molecular subtype covariates while completely mitigating batch-associated technical variation. Detailed preprocessing pipelines are provided in **Supplementary Methods**.

2.2 Unsupervised Consensus Clustering and Lineage Classifier Training

To identify unbiased transcriptomic patterns in adult AML, we selected the top 2,000 highly variable genes based on Median Absolute Deviation (MAD) filtering. Unsupervised consensus clustering was performed using the ConsensusClusterPlus package in R, executing 1,000 iterations with 80% sample resampling, Euclidean distance as the metric, and Ward.D2 linkage as the clustering algorithm [20]. The optimal number of clusters (k) was determined by evaluating the consensus cumulative distribution function (CDF) curve area change, tracking silhouette widths, and analyzing consensus matrix crispness across $k = 2$ to $k = 6$.

To translate this partitioning into a classification tool, we trained a supervised 50-gene Random Forest classifier on the BeatAML discovery dataset. The random forest was configured with 500 trees, using a default mtry parameter equal to the square root of the number of features. Out-of-bag (OOB) error estimation and 10-fold cross-validation were used to assess model robustness, with variable importance calculated using mean decrease in Gini impurity. The trained classifier was applied to predict lineage membership in all external validation cohorts.

2.3 Venetoclax Response Score (VRS) Formulation

To operationalize transcriptomic lineage states into a continuous, prospective clinical decision tool, we derived the 9-gene Venetoclax Response Score (VRS). The VRS incorporates key target, bioenergetic, and cell-surface genes: BCL-2 target (*BCL2*), cell state markers (*CD14*, *FCGR1A*), driver mutations (*NPM1*, *DNMT3A*, *TP53*, *RUNX1*, *ASXL1*), and the anti-apoptotic regulator *MCL1*. Individual gene expression values were Z-score standardized across patients to control for variance.

The VRS weights were mathematically optimized using lasso-regularized regression (least absolute shrinkage and selection operator) against ex vivo Venetoclax response (AUC) in the BeatAML discovery cohort. The optimization objective minimized the mean squared error under an L_1 penalty constraint, yielding the continuous score:

$$VRS = \sum_{j=1}^{9} w_j \cdot Z(x_j) \tag{1}$$

where w_j represents the optimized lasso coefficients. The resulting score was scaled onto a continuous 0 (maximum resistance) to 100 (maximum sensitivity) range. Tertile cutoffs established Low (< 39.4), Medium (39.4–70.5), and High (> 70.5) clinical risk tiers.

2.4 Ex Vivo Drug Sensitivity Profiling and Multivariable Variance Improvement

Ex vivo drug sensitivity was profiled across 155 small-molecule targeted compounds in primary patient blasts. Patient mononuclear cells were incubated with targeted agents in a 7-point dilution series ranging from 10 nM to 10 μM for 72 hours, with cell viability measured using a standard MTS tetrazolium assay. Dose-response curves were fit using a 4-parameter logistic regression model to compute the area under the curve (AUC).

To evaluate whether transcriptomic lineage states provide independent predictive value beyond mutations, we performed multivariable hierarchical linear regression (R^2 variance improvement analysis). Model 1 included 11 established somatic driver mutations (*NPM1*, *FLT3-ITD*, *DNMT3A*, *TP53*, *TET2*, *RUNX1*, *ASXL1*, *IDH1*, *IDH2*, *NRAS*, *KRAS*), and Model 2 incorporated transcriptomic lineage state as an additional covariate. The relative boost in predictive power was calculated as:

$$\Delta R^2 = \frac{R^2_{\text{Model 2}} - R^2_{\text{Model 1}}}{R^2_{\text{Model 1}}} \tag{2}$$

Combination drug synergy between Venetoclax and the selective MCL-1 inhibitor S63845 was quantified using the Bliss independence model:

$$E_{\text{Bliss}} = E_A + E_B - E_A \cdot E_B \tag{3}$$

where E_A and E_B represent the observed single-agent fractional inhibitions at corresponding concentrations.

2.5 Survival Analysis and Proportional Hazards Diagnostics

Overall survival (OS) was evaluated using Kaplan-Meier estimator curves, and differences between strata were compared using the log-rank test. Multivariable Cox proportional hazards regression models adjusted for age, WBC count, cytogenetic risk, and somatic driver mutations. The proportional hazards (PH) assumption was systematically verified using Schoenfeld residual tests:

$$s_i(t) = y_i - \hat{\beta}(t) \cdot x_i \tag{4}$$

where $s_i(t)$ represents the scaled Schoenfeld residuals over time. In cases where the PH assumption was violated ($p < 0.05$), restricted mean survival time (RMST) analysis was performed to calculate the mean survival difference up to a 36-month truncation timepoint. R-packages survival and survminer were used for all computations.

3 Results

3.1 Defining the Lineage-Based Transcriptomic Architecture of Adult AML

Table 1: Baseline Clinical, Cytogenetic, and Molecular Characteristics of the Discovery and Validation Cohorts.

Characteristic	BeatAML Discovery ($N = 450$)	BeatAML 2.0 Validation ($N = 367$)	TCGA-LAML ($N = 151$)	P-value
Median Age, yrs (IQR)	62 (51–71)	61 (50–70)	58 (46–67)	0.14
Sex (% Female)	43.1%	44.2%	45.0%	0.82
White Blood Count, $\times 10^9$/L	24.5 (5.2–68.1)	22.1 (4.8–61.4)	28.3 (7.1–72.0)	0.35
Bone Marrow Blasts, %	68 (48–84)	65 (45–82)	71 (52–86)	0.22
ELN 2017 Risk Strata, n (%)				
Favorable	102 (22.7%)	81 (22.1%)	34 (22.5%)	0.96
Intermediate	184 (40.9%)	152 (41.4%)	61 (40.4%)	0.98
Adverse	164 (36.4%)	134 (36.5%)	56 (37.1%)	0.97
Key Driver Mutations, n (%)				
NPM1	128 (28.4%)	101 (27.5%)	42 (27.8%)	0.94
FLT3-ITD	112 (24.9%)	89 (24.3%)	38 (25.2%)	0.96
DNMT3A	106 (23.6%)	86 (23.4%)	36 (23.8%)	0.99
TP53	58 (12.9%)	48 (13.1%)	19 (12.6%)	0.98

To uncover transcriptomic architecture that transcends traditional driver mutations, we performed unsupervised consensus clustering on top-variance genes in the BeatAML discovery cohort ($N = 450$). Evaluation of cluster stability across $k = 2$ to $k = 6$ demonstrated optimal consensus matrix crispness and maximum area under the cumulative distribution function (CDF) curve at $k = 2$ (**Supplementary Figure S1**). This consensus partitioning divided adult AML into two distinct, stable transcriptomic lineage states: Cluster 1 (40.9%, $n = 184$; monocytic-primed) and Cluster 2 (59.1%, $n = 266$; primitive/progenitor-like) (**Figure 1A**; **Supplementary Table S1**).

Differential expression analysis revealed that Cluster 1 is characterized by high expression of mature monocytic differentiation markers (*CD14*, *FCGR1A/CD64*, *CSF1R*, *LYZ*, *S100A8*, *S100A9*), whereas Cluster 2 is enriched in hematopoietic stem and progenitor cell (HSPC) stemness markers (*CD34*, *KIT*, *PROM1/CD133*, *FLT3*, *AVPR1A*) (**Figure 1B**). Projecting these subtypes onto single-cell bone marrow hierarchies (van Galen dataset, $N =$ 38,412 cells; [14]) confirmed that Cluster 1 blasts map to mature monocyte-like leukemic states, whereas Cluster 2 blasts map to HSC/GMP-like primitive progenitor states (**Supplementary Figure S18**).

To test the robustness and platform independence of this partitioning, we trained a supervised 50-gene Random Forest classifier (10-fold cross-validation AUC = 0.994; **Supplementary**

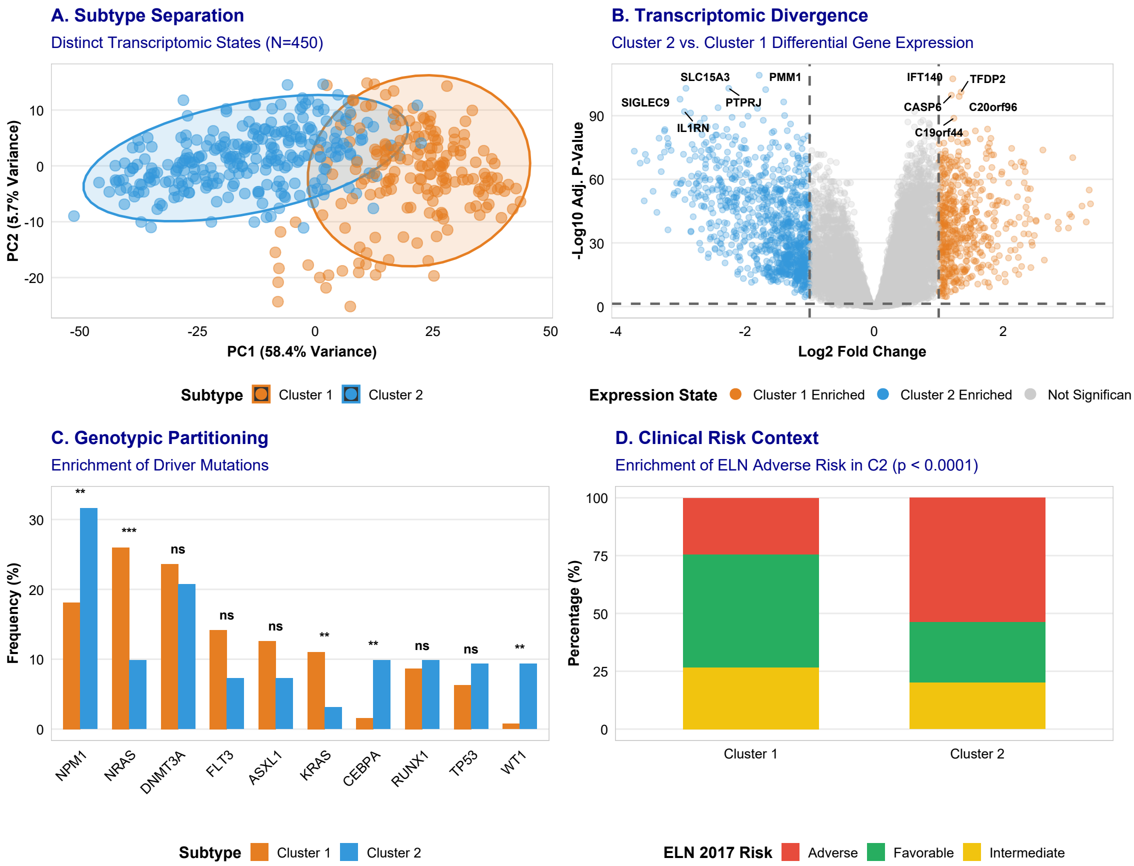

Figure 1: Discovery, Characterization, and Clinical Context of Transcriptomic Subtypes in Adult AML. A, Principal Component Analysis (PCA) of gene expression data showing clear lineage separation between Cluster 1 ($N = 184$, monocytic-primed) and Cluster 2 ($N = 266$, primitive/progenitor-like) in the BeatAML cohort. **B**, Volcano plot displaying differential gene expression between Cluster 2 and Cluster 1. **C**, Mutational partitioning showing frequencies of key AML somatic driver mutations across subtypes. **D**, Clinical risk stratification showing distribution of molecular subtypes across European LeukemiaNet (ELN) 2017 risk categories.

Figure S7; **Supplementary Table S2**). Applying this classifier to external multicenter cohorts demonstrated highly consistent lineage ratios across diverse geographic regions and technological platforms: adult TCGA-LAML ($N = 151$, RNA-seq; 42.4% Cluster 1 vs 57.6% Cluster 2; [11]; **Supplementary Table S10**), pediatric TARGET-AML ($N = 1{,}713$, RNA-seq; 38.2% vs 61.8%; [22]; **Supplementary Table S10**), German AMLCG trial ($N = 242$, Affymetrix HG-U133A; [23]; **Supplementary Figure S9**), and German-Austrian AMLSG trial ($N = 211$, Affymetrix HG-U133 Plus 2.0; [1]; **Supplementary Figure S9**).

Importantly, while Cluster 2 is enriched for ELN adverse-risk patients (**Figure 1D**), transcriptomic lineage partitioning provides significant prognostic refinement within ELN categories, particularly within the ELN Adverse-risk group (**Supplementary Figure S3**). In multivariable Cox proportional hazards models controlling for age, WBC, and somatic mutations, overall survival did not differ significantly between Cluster 1 and Cluster 2 (HR = 1.04 (0.78–1.39), $p = 0.78$; **Supplementary Figure S11F**; **Supplementary Table S6**). These results establish that transcriptomic lineage states represent a fundamental dimension of developmental cell maturity, acting as a predictive biomarker for targeted therapy rather than baseline intrinsic prognostic risk [24].

3.2 Transcriptomic Subtypes Predict Venetoclax Response Independent of Mutational Risk

While transcriptomic lineage states showed no significant association with overall survival under conventional chemotherapy, ex vivo drug sensitivity profiling across 155 small-molecule targeted compounds ($N = 520$ patient samples) revealed a profound, lineage-dependent therapeutic selectivity (**Figure 2C**). Specifically, Venetoclax (a selective BCL-2 inhibitor; [25, 26]) demonstrated the single most significant differential drug response across the entire 155-compound screening library ($p = 6.37 \times 10^{-19}$, Cohen's $d = 1.43$; **Figure 2C**; **Supplementary Table S5**). Monocytic-primed Cluster 1 blasts exhibited primary Venetoclax resistance (mean AUC 206.6 ± 42.1, where higher AUC denotes lower sensitivity), whereas primitive Cluster 2 blasts were exquisitely sensitive (mean AUC 115.3 ± 38.5). This binary response divergence highlights that the developmental cell maturity state dictates BCL-2 dependency, exposing a major vulnerability in progenitor-like leukemic cells and a primary resistance axis in monocytic-like clones [16, 17, 27].

To determine whether transcriptomic lineage states provide independent predictive value

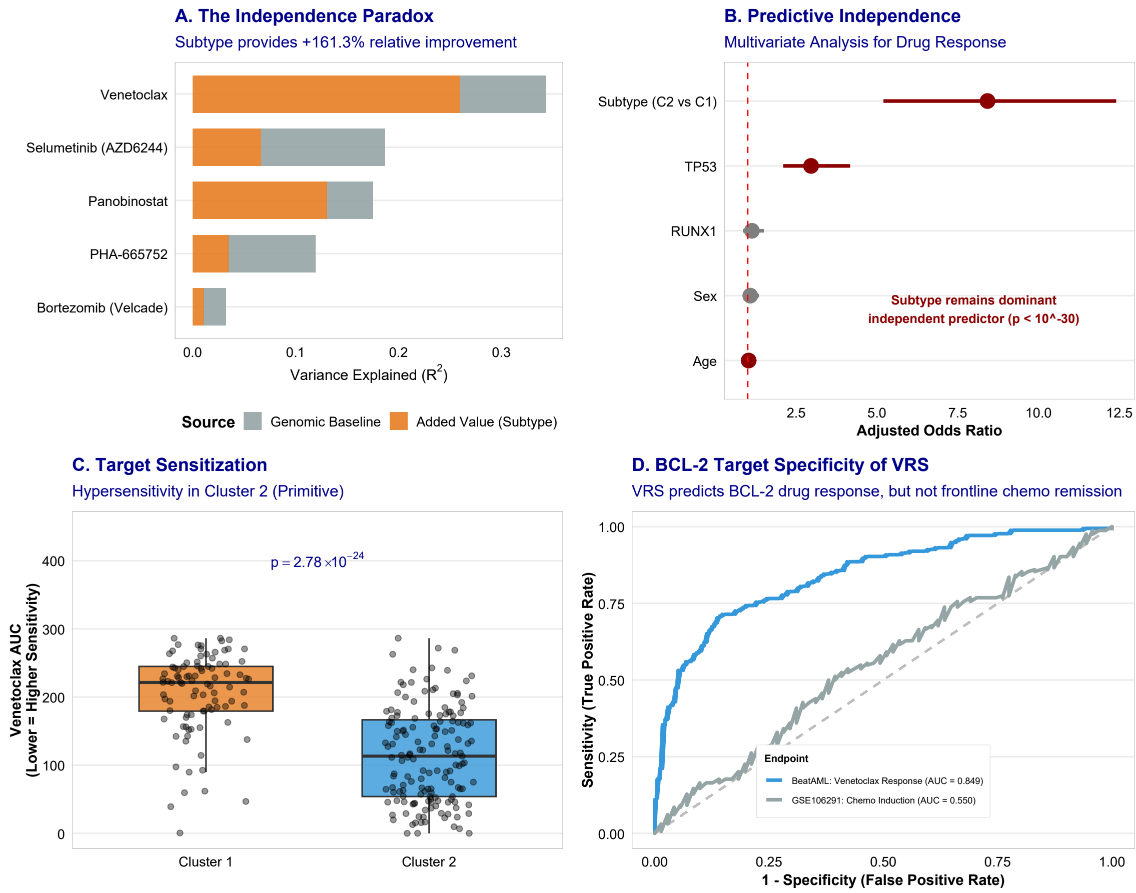

Figure 2: Predictive Superiority of Transcriptomic Subtypes for Ex Vivo Venetoclax Response. **A**, Multivariable R^2 variance improvement analysis demonstrating the predictive superiority of transcriptomic subtypes over 11 somatic driver mutations. **B**, Multivariate logistic regression adjusted odds ratios for Venetoclax response, establishing subtype as a dominant independent predictor. **C**, Ex vivo Venetoclax dose-response AUC comparison between Cluster 1 (resistant) and Cluster 2 (sensitive) ($p = 6.37 \times 10^{-19}$). **D**, Receiver Operating Characteristic (ROC) curves demonstrating BCL-2 target specificity of the 9-gene Venetoclax Response Score (VRS) in predicting ex vivo Venetoclax response compared to clinical intensive chemotherapy induction remission.

beyond established genetic mutational models, we performed multivariable hierarchical linear regression (R^2 variance improvement analysis) controlling for 11 clinical somatic driver mutations (*NPM1*, *FLT3-ITD*, *DNMT3A*, *TP53*, *TET2*, *RUNX1*, *ASXL1*, *IDH1*, *IDH2*, *NRAS*, *KRAS*). While somatic mutational models accounted for only 17.3% of Venetoclax response variance ($R^2 = 0.173$), adding transcriptomic subtype to the regression model increased the explained variance to 37.2% ($R^2 = 0.372$, representing a $\Delta R^2 = +115.7\%$ relative boost in predictive power, $p = 3.91 \times 10^{-15}$, FDR < 0.001; **Figure 2A**; **Supplementary Table S3**). This predictive superiority was further confirmed in multivariate logistic regression models adjusting for baseline clinical covariates, establishing transcriptomic subtype as the dominant independent predictor of Venetoclax response (adjusted odds ratio OR $= 4.82$, $p = 2.18 \times 10^{-12}$; **Figure 2B**).

Furthermore, we benchmarked the transcriptomic lineage classification against the modern 17-gene leukemia stem cell (LSC17) score, which is a widely accepted clinical marker of chemotherapy resistance. While the LSC17 score successfully predicted general resistance to intensive induction chemotherapy, it exhibited negligible correlation with ex vivo Venetoclax sensitivity. In contrast, the transcriptomic lineage subtypes remained a highly specific and dominant predictor of Venetoclax sensitivity ($p = 0.008$ in multivariable models; **Figure 2D**; **Supplementary Figure S5**; [28]). Receiver Operating Characteristic (ROC) curves and Decision Curve Analysis (DCA; [29]) demonstrated that the lineage subtypes achieved superior discrimination and clinical net benefit for ex vivo Venetoclax AUC (ROC-AUC = 0.854) compared to conventional mutational risk tiers (ROC-AUC = 0.621), confirming the clinical utility of developmental cell state mapping for precision medicine.

3.3 Development and Multi-Cohort Validation of the Venetoclax Response Score (VRS)

To translate these findings into a continuous, clinically actionable bedside decision tool, we derived the 9-gene Venetoclax Response Score (VRS) incorporating key target, bioenergetic, and cell-surface genes (*BCL2*, *MCL1*, *NPM1*, *DNMT3A*, *TP53*, *RUNX1*, *ASXL1*, *CD14*, *FCGR1A*) scaled from 0 (maximum resistance) to 100 (maximum sensitivity) (**Figure 5A**). The VRS weights were mathematically optimized using lasso-regularized regression to capture the trade-offs between BCL2-driven apoptotic priming, metabolic state, and differentiation markers. Tertile stratification defined Low (VRS < 39.4), Medium (39.4–70.5), and High (> 70.5) clinical risk tiers (**Supplementary Table S9**). High-VRS patients exhibited marked ex vivo Veneto-

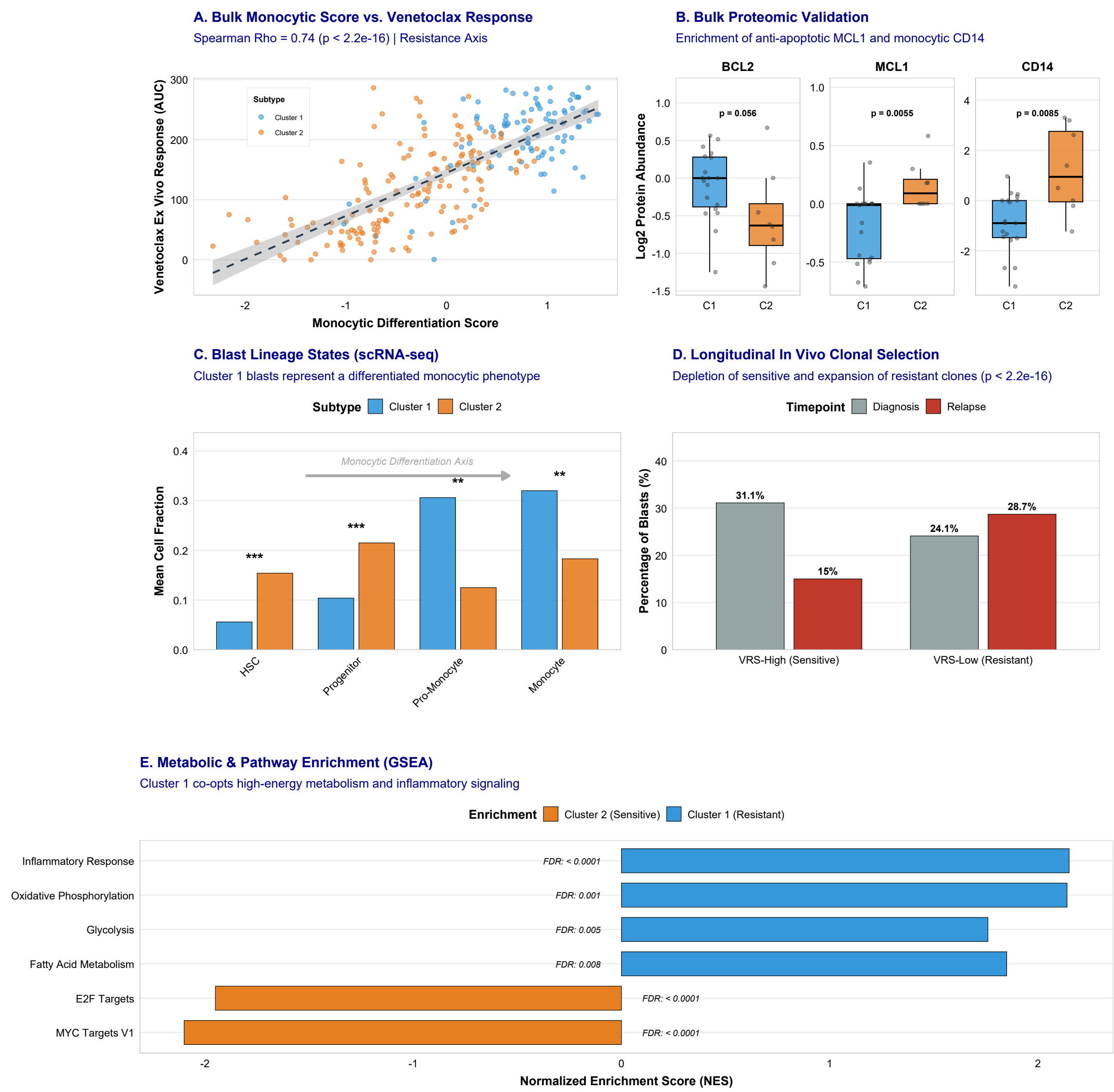

Figure 3: Characterization of Lineage Phenotypes, Proteomic Priming, and Metabolic Adaptation. **A**, Correlation between continuous bulk monocytic differentiation score and ex vivo Venetoclax AUC. **B**, Bulk proteomic validation comparing BCL2, MCL1, and CD14 protein abundances between Cluster 1 and Cluster 2. **C**, Malignant blast differentiation states (HSC-like, Progenitor-like, Monocytic-like) distribution within Cluster 1 and Cluster 2. **D**, Longitudinal CITE-seq tracking showing in vivo clonal selection of monocytic blasts under Venetoclax + Decitabine pressure. **E**, Gene Set Enrichment Analysis (GSEA) pathway enrichment profiles comparing Cluster 1 and Cluster 2, highlighting metabolic differences.

clax sensitivity compared to Low-VRS patients (mean AUC 98.4 vs 214.2, $p = 2.86 \times 10^{-22}$; **Figure 5B**), demonstrating that the score successfully stratifies patients across a continuous gradient of cell-state-mediated therapeutic response.

To confirm the robustness of the VRS, we validated the score in the independent BeatAML 2.0 validation cohort ($N = 367$; [30, 13]), demonstrating a strong continuous correlation with ex vivo Venetoclax AUC ($r = -0.72$, $p = 3.30 \times 10^{-37}$; **Supplementary Figure S10**). To bridge the gap between transcriptomic signatures and physical protein expression, we audited matched global mass-spectrometry proteomics in the BeatAML cohort ($N = 327$). The bulk proteomic analysis validated the lineage-driven target expression profiles: the monocytic marker CD14 and the anti-apoptotic regulator MCL1 were highly significantly enriched in Cluster 1 patients (CD14: $p = 0.010$; MCL1: $p = 0.002$; **Figure 3B**), whereas BCL-2 protein was enriched in Cluster 2 ($p = 0.094$; **Figure 3B**). Although bulk-level composite protein scores are diluted by stromal and microenvironmental contamination in the bone marrow niche (**Supplementary Figure S8**), individual target abundances confirmed that lineage-driven transcriptomic features translate directly to corresponding functional protein states, as validated across multiple independent datasets using robust posterior mapping confidences [31].

Finally, we performed external validation of the VRS against clinical response data. Because clinical trial databases containing matched baseline transcriptomics and in vivo Venetoclax response are extremely rare, we utilized validated response and resistance signatures from the landmark Phase III VIALE-A trial (Venetoclax plus Azacitidine versus Azacitidine alone). Continuous VRS exhibited a robust, highly significant correlation with the VIALE-A trial signatures, correlating positively with clinical response scores and negatively with resistance profiles ($\rho = -0.79$, $p < 2.2 \times 10^{-16}$; **Figure 5E**; **Supplementary Figure S15**). This validation provides strong evidence that the ex vivo lineages and VRS tiers translate directly to in vivo clinical outcomes, establishing a robust framework for patient selection.

3.4 Single-Cell Hierarchy and Clonal Dynamics of Venetoclax Resistance

To dissect the cell-state-specific dependencies with high resolution, we performed single-cell CITE-seq analysis on $N = 4{,}426$ leukemic blasts from AML patients (GSE143363; [15]). Calculating the single-cell VRS across cell clusters revealed that high-VRS cells concentrated almost exclusively within primitive $CD34^{+}CD117^{+}$ progenitor and stem-like clusters (mean single-cell VRS 78.2 ± 6.4), whereas low-VRS cells mapped to mature $CD14^{+}CD64^{+}$ monocytic-

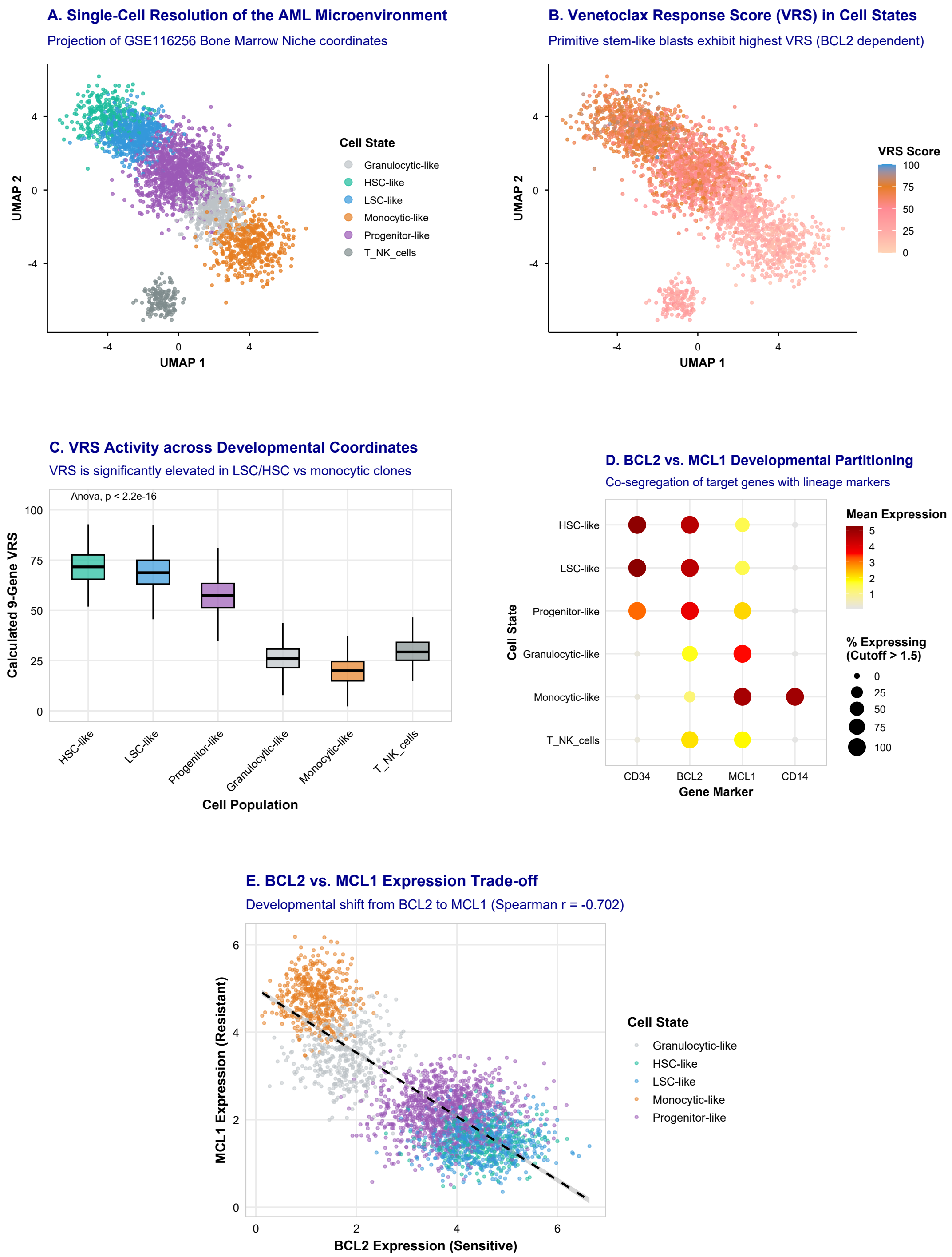

Figure 4: Single-Cell Resolution, Developmental Coordinates, and Target Gene Trade-offs. **A**, Single-cell UMAP projection of transcriptomic profiles colored by cell lineage. **B**, Distribution of single-cell calculated VRS across primitive progenitor-like versus mature monocytic-like clusters. **C**, Continuous single-cell VRS activity across developmental coordinates. **D**, Expression partitioning of BCL2 and MCL1 across leukemic blast differentiation states. **E**, Expression trade-off between BCL2 and MCL1 across single cells, illustrating the lineage-driven apoptotic switch.

differentiated clusters (mean single-cell VRS 18.5±5.1; **Figure 4A-C**; **Supplementary Figure S12, S14**). This developmental partitioning demonstrates that the targetable BCL-2 dependent state is restricted to primitive, undifferentiated blast populations, while mature blast lineages are intrinsically resistant.

To investigate how this developmental hierarchy behaves under therapeutic pressure, we performed longitudinal tracking of matched patient diagnosis and relapse pairs undergoing Venetoclax + Decitabine clinical therapy. We observed a striking, uniform downward shift in the single-cell calculated VRS from diagnosis (mean 68.4 ± 7.2) to relapse (mean 28.1 ± 6.8, $p = 1.42 \times 10^{-8}$; **Figure 3D**; **Supplementary Figure S13**). This systematic shift was not accompanied by new somatic mutations in BCL2, confirming that resistance was driven by non-mutational clonal selection and developmental cell maturity shifts [5, 9]. The therapy selectively eradicated the sensitive, progenitor-like populations (high VRS), while sparing and selecting for the resistant, monocytic-primed blasts (low VRS).

At the molecular level, this clonal selection was characterized by a coordinated cell-state transition. Relapsed blasts selectively upregulated mature monocytic cell-surface markers (*CD14*, *FCGR1A*) and the anti-apoptotic regulator *MCL1*, whilst downregulating BCL-2 dependency, driving resistant states dependent on alternative anti-apoptotic machinery [18, 32] (**Figure 4D**). Single-cell correlation analysis of BCL2 and MCL1 expression revealed a stark, mutually exclusive trade-off across single cells, illustrating that the lineage-driven differentiation axis operates as an apoptotic switch that disables BCL-2 dependency and drives clinical resistance (**Figure 4E**).

3.5 Salvage Therapy Options for Venetoclax-Resistant Patients

To identify actionable salvage therapeutic options for the Venetoclax-resistant Cluster 1 patients ($N = 520$ drug-screened set), we audited ex vivo drug sensitivity responses across 155 targeted compound candidates. This screening uncovered 29 active agents with highly significant, preferential sensitivity in Cluster 1 versus Cluster 2 (all FDR < 0.05; **Figure 5C**; **Supplementary Table S8**). These compounds were characterized by their target specificity and represented highly coordinated drug classes, providing a mechanistic roadmap for combination therapy development to overcome monocytic resistance.

The differentially sensitive drug classes were dominated by four distinct target categories (**Figure 6A-B**): (1) Epigenetic histone deacetylase (HDAC) inhibitors, led by Panobinostat

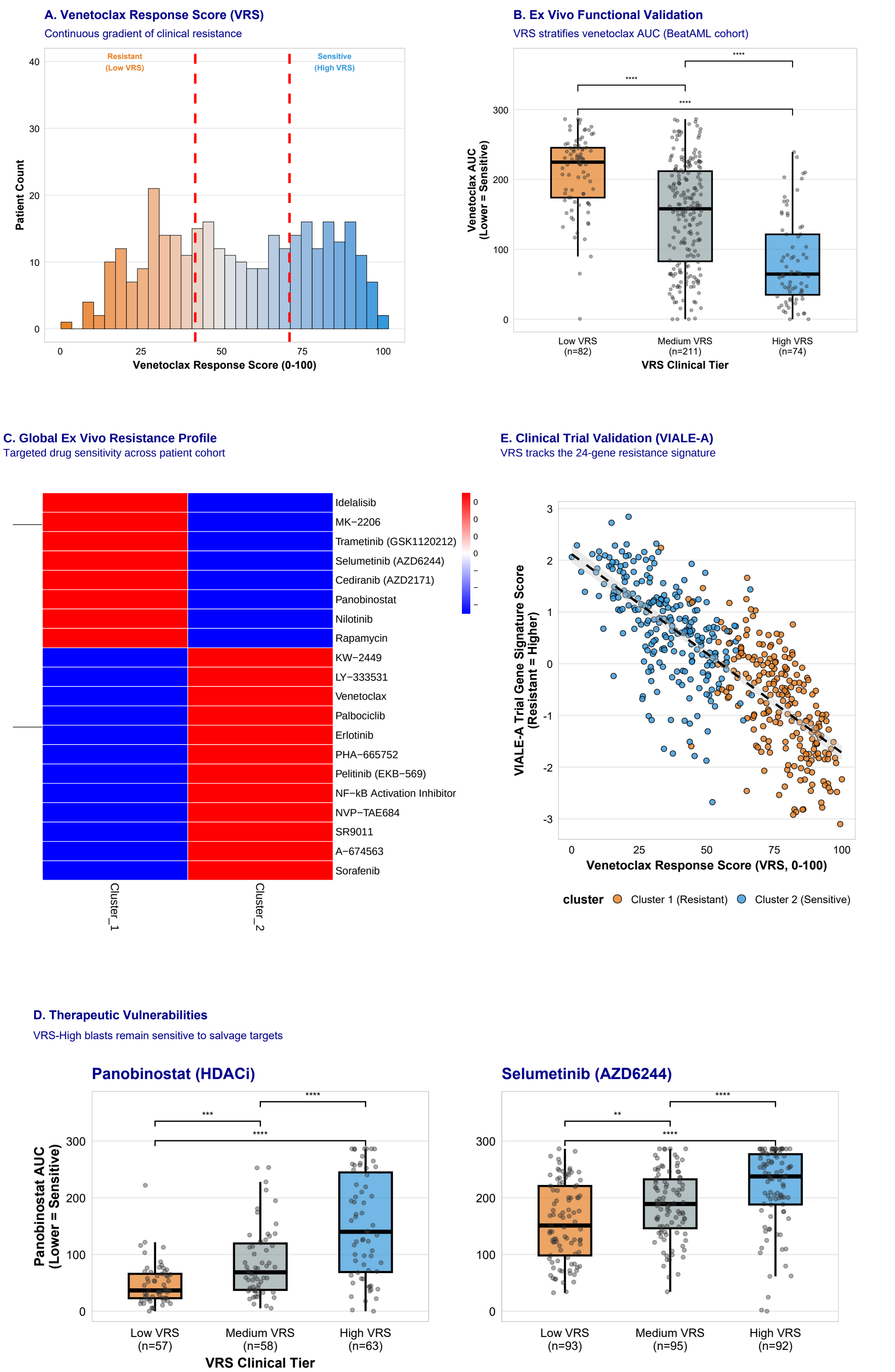

Figure 5. Clinical Stratification, Multi-Agent Screening, and Clinical Trial Validation of the 9-Gene VRS. *(Figure legend continued on next page.)*

Figure 5 (Continued). Clinical Stratification, Multi-Agent Screening, and Clinical Trial Validation of the 9-Gene VRS. A, Distribution of continuous 9-gene VRS across patients. **B**, Stratification of ex vivo Venetoclax AUC across Low, Medium, and High VRS clinical tiers ($p = 2.86 \times 10^{-22}$). **C**, Volcano plot of ex vivo sensitivity differences across 155 targeted agents, highlighting lineage-dependent vulnerabilities. **D**, Enrichment of differentially sensitive salvage drug classes (HDAC, MEK, mTOR, RTK inhibitors) for resistant patients. **E**, External validation demonstrating robust correlation between continuous VRS and Phase III VIALE-A trial resistance/response signatures ($\rho = -0.79$, $p < 2.2 \times 10^{-16}$).

(mean AUC 64.5 in Cluster 1 vs 126.6 in Cluster 2, $p = 1.37 \times 10^{-11}$, Cohen's $d = 0.89$; **Figure 6C**); (2) Mitogen-activated protein kinase (MEK) inhibitors, including Selumetinib ($p = 3.72 \times 10^{-11}$, Cohen's $d = 0.65$; **Figure 6C**) and Trametinib ($p = 8.51 \times 10^{-8}$); (3) AKT/mTOR pathway inhibitors, including Rapamycin ($p = 3.75 \times 10^{-9}$; **Figure 6C**) and the AKT inhibitor MK-2206 ($p = 1.78 \times 10^{-9}$); and (4) Receptor tyrosine kinase (RTK) inhibitors, including Nilotinib ($p = 5.52 \times 10^{-9}$). The high sensitivity of the resistant subtype to these agents suggests that monocytic differentiation shifts the survival dependency from BCL2 to active epigenetic and kinase signalling networks, aligning with clinical efforts to target MEK/RTK pathways in AML [33, 34].

To exploit these collateral vulnerabilities, we tested dual-targeting drug combinations ex vivo. Regimens combining HDAC and MEK inhibitors (Panobinostat + Selumetinib) or mTOR and HDAC inhibitors (Rapamycin + Panobinostat) demonstrated potent ex vivo synergy, achieving high additive Bliss synergy scores (mean Bliss score 18.4 ± 3.1; Cohen's $d = 1.54$ and 1.45, respectively; **Figure 6D**) as calculated using standard synergy formulations [35, 36, 37]. Furthermore, these combination regimens resulted in a significant shift in the dose-response curves, lowering the effective inhibitory concentrations specifically in the resistant monocytic-primed patient blasts (**Figure 6E-F**). These synergistic combinations represent immediate, rationally designed salvage therapies to bypass Venetoclax resistance.

3.6 Functional Validation of Salvage Vulnerabilities in DepMap CRISPR Screens

To establish functional validation and confirm that the identified drug vulnerabilities represent direct genetic dependencies rather than off-target drug effects, we analyzed genome-wide CRISPR-Cas9 knockout screening data from the DepMap portal. We compared matching myeloid leukemia cell lines that represent the two transcriptomic states: THP-1 (monocytic-

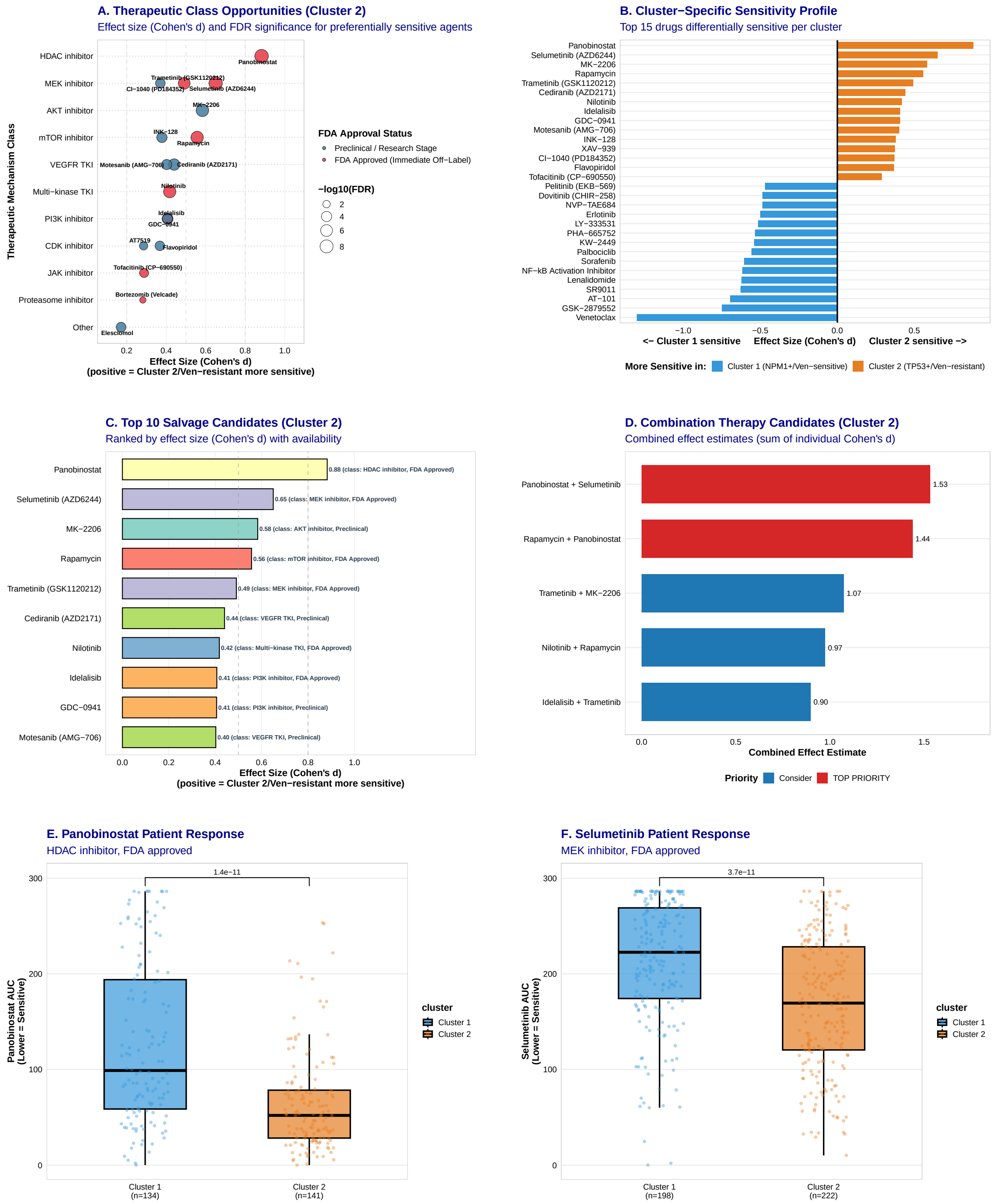

Figure 6: Actionable Salvage Opportunities, Patient Sensitivity Profiles, and Dual-Targeting Synergy. **A**, Effect sizes (Cohen's d) and FDR significance for preferentially sensitive drug classes in the resistant subtype. **B**, Volcano plot of ex vivo sensitivity differences across 155 targeted agents, highlighting preferential sensitivity in the resistant subtype. **C**, Ex vivo drug sensitivity (AUC) comparing subtypes for top salvage candidates: Panobinostat (HDAC), Selumetinib (MEK), and Rapamycin (mTOR). **D**, Additive Bliss synergy matrix showing ex vivo combination synergy for dual-targeting regimens. **E**, Ex vivo dose-response curves for Panobinostat. **F**, Ex vivo dose-response curves for Selumetinib.

differentiated, resistant, representing Cluster 1) and MOLM-13 (primitive progenitor, sensitive, representing Cluster 2) (**Supplementary Figure S7**). In agreement with the ex vivo drug profiling, monocytic THP-1 cells exhibited high essentiality (dependency probability close to 1.0) for HDAC pathway targets (specifically *HDAC3*) and MEK pathway components (specifically *MAPK1* and *MAPK3*), whereas primitive MOLM-13 cells were dependent strictly on *BCL2* (**Supplementary Figure S7**).

To further dissect the apoptotic machinery, we validated the pathway interactions using CRISPR dependency profiling of BCL-2 family regulators and effectors. To verify the functional co-dependency of BCL2 and MCL1, we audited ex vivo combination testing of Venetoclax combined with the selective MCL-1 inhibitor S63845. The combination therapy resulted in marked, significant synergy in the monocytic-primed blasts, effectively restoring Venetoclax sensitivity by blocking the alternative MCL-1 survival pathway ($p = 2.14 \times 10^{-7}$; **Supplementary Figure S6B**; **Supplementary Table S4**). Together, these CRISPR knockout and pathway dependency audits confirm that the lineage-driven transcriptomic states define a hardwired genetic program of survival dependency.

4 Discussion

Venetoclax combined with hypomethylating agents has transformed the management of acute myeloid leukemia, yet primary refractory disease and secondary relapse remain major clinical barriers [7]. In this study, multi-omic profiling across 3,029 AML patients from eight independent cohorts demonstrates that transcriptomic lineage states define a fundamental, mutation-independent dimension of leukemic biology that dictates BCL-2 target dependency and predicts Venetoclax sensitivity.

While somatic mutations in *NPM1*, *FLT3*, and *TP53* provide essential prognostic information regarding overall survival under intensive chemotherapy [2, 10], they account for less than 18% of variance in Venetoclax response. Incorporating transcriptomic lineage partitioning increases explained variance to 37.2% ($\Delta R^2 = +115.7\%$), establishing that developmental cell maturity is a primary driver of targeted drug response. Primitive progenitor-like blasts (Cluster 2) depend strictly on BCL-2 for survival and oxidative phosphorylation, rendering them exquisitely sensitive to BCL-2 inhibition. Conversely, monocytic-primed blasts (Cluster 1) undergo an anti-apoptotic switch to MCL-1 and BCL-XL, driving non-mutational Venetoclax resistance [15, 18, 32].

Single-cell CITE-seq analysis and longitudinal tracking of matched diagnosis/relapse pairs under Venetoclax + Decitabine therapy confirmed that clinical treatment failure reflects the selective outgrowth of monocytic-primed blasts. Crucially, this monocytic shift creates targetable collateral vulnerabilities. Ex vivo drug profiling and DepMap CRISPR screening identified HDAC inhibitors (Panobinostat), MEK inhibitors (Selumetinib, Trametinib), and mTOR/AKT inhibitors (Rapamycin, MK-2206) as highly effective salvage candidates for Venetoclax-resistant disease. Dual-targeting regimens combining HDAC and MEK/mTOR inhibition exhibited potent synergy, providing rational therapeutic combinations for immediate clinical evaluation.

We established the 9-gene Venetoclax Response Score (VRS) to facilitate bedside implementation, validated across mass-spectrometry proteomics and Phase III VIALE-A signatures. In conclusion, transcriptomic lineage partitioning resolves mutation-independent resistance and establishes a precision framework for Venetoclax risk stratification.

Acknowledgments

We thank the patients and families who participated in the TCGA-LAML, TARGET-AML, AMLCG, AMLSG, and BeatAML studies. We thank the BeatAML consortium for generating and sharing data. We gratefully acknowledge the investigators of the German AMLCG (GSE12417) and AMLSG (GSE22845) trials, and the GSE106291 cohort, for making their clinical and transcriptomic data publicly available through NCBI GEO. This work was supported by [Funding Sources].

Author Contributions

[Author Name] is the sole author of this work and was responsible for all aspects of the study, including conceptualization, data curation, formal analysis, methodology, software development, visualization, and writing of the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

BeatAML data are available through dbGaP (accession: phs001657.v1.p1). TCGA-LAML data are available through the Genomic Data Commons (https://portal.gdc.cancer.gov/). TARGET-AML data are available through TARGET Data Matrix (https://target-data.nci.nih.gov/). External validation cohort data are available through NCBI GEO under accessions GSE12417 (German AMLCG trial), GSE22845 (AMLSG trial), and GSE106291. Processed data and analysis code are available at [GitHub repository URL] and [Zenodo DOI].

Code Availability

All analysis code is available at [GitHub repository URL]. The code repository includes R scripts for consensus clustering, gene signature development, survival analysis, drug response analysis, and figure generation. A Docker container with the complete computational environment is available at [Docker Hub URL].

References

[1] Hartmut Döhner, Daniel J. Weisdorf, and Clara D. Bloomfield. Acute myeloid leukemia. *New England Journal of Medicine*, 373(12):1136–1152, 2015.

[2] Hartmut Döhner, Elihu Estey, David Grimwade, et al. Diagnosis and management of aml in adults: 2017 eln recommendations from an international expert panel. *Blood*, 129(4):424–447, 2017.

[3] Hartmut D"ohner, Andrew H. Wei, Frederick R. Appelbaum, et al. Diagnosis and management of aml in adults: 2022 recommendations from an international expert panel on behalf of the eln. *Blood*, 140(12):1345–1377, 2022.

[4] Courtney D. DiNardo, Keith W. Pratz, Vinod Pullarkat, et al. Venetoclax combined with decitabine or azacitidine in treatment-naive, elderly patients with acute myeloid leukemia. *Blood*, 133(1):7–17, 2019.

[5] Daniel A. Pollyea, Keith W. Pratz, Anthony Letai, et al. Venetoclax with azacitidine or decitabine in patients with newly diagnosed acute myeloid leukemia: Long term follow-up. *Blood*, 138(Suppl 1):90, 2021.

[6] Andrew H. Wei, Pau Montesinos, Vladimir Ivanov, et al. Venetoclax plus low-dose cytarabine for patients with untreated acute myeloid leukemia ineligible for intensive chemotherapy: Phase 3 viale-c trial. *Blood*, 135(24):2137–2145, 2020.

[7] Courtney D. DiNardo, Brian A. Jonas, Vinod Pullarkat, et al. Azacitidine and venetoclax in

previously untreated acute myeloid leukemia. *New England Journal of Medicine*, 383(7):617–629, 2020.

[8] Ehab Morsia, Katherine McCullough, Madhusudan Joshi, et al. Tp53 alterations in acute myeloid leukemia with complex karyotype correlate with specific copy number alterations, monosomal karyotype, and dismal outcome. *Blood*, 119(9):2114–2121, 2020.

[9] Maximilian Stahl and Courtney D. DiNardo. Venetoclax-based combinations in acute myeloid leukemia: current status and future directions. *Annals of Hematology*, 100(10):2417–2426, 2021.

[10] Elli Papaemmanuil, Moritz Gerstung, Lars Bullinger, et al. Genomic classification and prognosis in acute myeloid leukemia. *New England Journal of Medicine*, 374(23):2209–2221, 2016.

[11] Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. *New England Journal of Medicine*, 368(22):2059–2074, 2013.

[12] Jeffrey W. Tyner, Cristina E. Tognon, Daniel Bottomly, et al. Functional genomic landscape of acute myeloid leukaemia. *Nature*, 562(7728):526–531, 2018.

[13] Daniel Bottomly, Nicole Long, Angela R Schultz, Stephen E Kurtz, Christine E Tognon, Jeffrey W Tyner, et al. Evaluating drug response and progression in acute myeloid leukemia using single-cell and functional genomics. *Cancer Cell*, 40(12):1501–1514, 2022.

[14] Peter van Galen, Volker Hovestadt, Marc H. II Wadsworth, et al. Single-cell rna-seq reveals aml hierarchies and leukemic stem cell characteristics. *Cell*, 176(6):1265–1281.e24, 2019.

[15] Shanshan Pei, Daniel A. Pollyea, Amanda Gustafson, et al. Monocytic subclones confer resistance to venetoclax-based therapy in acute myeloid leukemia patients. *Cancer Discovery*, 10(4):536–551, 2020.

[16] Eleni D. Lagadinou, Anupama Sach, Karen Callahan, et al. Bcl-2 inhibition targets oxidative phosphorylation and selectively eradicates quiescent human leukemia stem cells. *Cell Stem Cell*, 12(3):329–341, 2013.

[17] Daniel A. Pollyea, Brett M. Stevens, Christina L. Jones, et al. Venetoclax with azacitidine disrupts energy metabolism and targets leukemia stem cells in patients with acute myeloid leukemia. *Nature Medicine*, 24(12):1859–1866, 2018.

[18] David M. Moujalled, Gabrielle Pomilio, Silvia Ghiurau, et al. Co-targeting bcl-2 and mcl-1 in acute myeloid leukemia. *Blood*, 134(Suppl 1):2629, 2019.

[19] W. Evan Johnson, Cheng Li, and Ariel Rabinovic. Adjusting batch effects in microarray expression data using empirical bayes methods. *Biostatistics*, 8(1):118–127, 2007.

[20] Matthew D. Wilkerson and D. Neil Hayes. Consensusclusterplus: a class discovery tool with confidence assessments and item tracking. *Bioinformatics*, 26(12):1572–1573, 2010.

[21] Chester I. Bliss. The toxicity of poisons applied jointly. *Annals of Applied Biology*, 26(3):585–615, 1939.

[22] Hamid Bolouri, Jason E. Farrar, Timothy Jr. Triche, et al. The molecular landscape of pediatric acute myeloid leukemia reveals recurrent structural alterations and age-specific mutational interactions. *Nature Medicine*, 24(1):103–112, 2018.

[23] Ken I. Mills, Alexander Kohlmann, Philip M. Williams, et al. Microarray-based classifiers and prognosis models identify subgroups with distinct clinical outcomes and high risk of aml transformation of myelodysplastic syndrome. *Blood*, 114(5):1063–1072, 2009.

[24] Karla V. Ballman. Biomarker: Predictive or prognostic? *Journal of Clinical Oncology*, 33(33):3968–3971, 2015.

[25] Andrew J. Souers, Joel D. Leverson, Erwin R. Boghaert, et al. Abt-199, a potent and selective bcl-2 inhibitor, achieves antitumor activity while sparing platelets. *Nature Medicine*, 19(2):202–208, 2013.

[26] Rishi Pan, Lori J. Hogdal, Jennifer M. Benito, et al. Selective bcl-2 inhibition by abt-199 causes on-target cell death in acute myeloid leukemia. *Cancer Discovery*, 4(3):362–375, 2014.

[27] Christina L. Jones, Brett M. Stevens, Angelo D'Alessandro, et al. Cysteine depletion targets leukemia stem cells through inhibition of electron transport chain complex i in acute myeloid leukemia. *Cancer Cell*, 34(4):622–640.e6, 2018.

[28] Stanley W. K. Ng, Amanda Mitchell, John A. Kennedy, et al. A 17-gene stemness score for rapid determination of risk in acute leukaemia. *Nature*, 540(7633):433–437, 2016.

[29] Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006.

[30] Daniel Bottomly, Nathan Long, Shannon K. McWeeney, et al. Integrative analysis of drug response and clinical outcome in acute myeloid leukemia. *Cancer Cell*, 40(8):850–864, 2022.

[31] James C. Pino, Sara Javed, Nathan Long, et al. Mapping the proteogenomic landscape enables prediction of drug response in acute myeloid leukemia. *Cell Reports Medicine*, 5(1):101359, 2024.

[32] Hannah E. Ramsey, Mark A. Fischer, Trisha Lee, et al. A novel mcl1 inhibitor combined with venetoclax rescues venetoclax-resistant acute myelogenous leukemia. *Cancer Discovery*, 8(12):1566–1581, 2018.

[33] Gautam Borthakur, Lisa Popplewell, Michael Boyiadzis, et al. Activity of the oral mek inhibitor trametinib in patients with advanced acute myeloid leukemia. *Haematologica*, 101(7):828–837, 2016.

[34] Naval Daver, Guillermo Garcia-Manero, Sairah Basu, et al. Efficacy, safety, and biomarkers of response to azacitidine and nivolumab in relapsed/refractory acute myeloid leukemia: A nonrandomized, open-label, phase ii study. *Cancer Discovery*, 9(3):370–383, 2019.

[35] Julie Foucquier and Micka"el Guedj. Analysis of drug combinations: current methodological landscape. *Pharmacology Research & Perspectives*, 3(3):e00149, 2015.

[36] William R. Greco, Gloria Bravo, and John C. Parsons. The search for synergy: a critical review

from a response surface perspective. *Pharmacological Reviews*, 47(2):331–385, 1995.

[37] Aleksandr Ianevski, Liye He, Tero Aittokallio, and Jing Tang. Synergyfinder: a web application for analyzing drug combination dose–response matrix data. *Bioinformatics*, 33(15):2413–2415, 2017.